

Tianyang Zhou

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Education

Carnegie Mellon University

Pittsburgh, PA

Ph.D. Student in Information Systems and Management

Aug. 2025 – June 2030 (Expected)

GPA: 3.95/4.00; **Coursework:** Generative AI, Optimization for ML, Advanced Topics in ML, Intermediate Statistics, Econometrics

Research Interest: LLM Agents, LLM Safety, LLM Reasoning & RL, Interpretable AI

Tsinghua University

Beijing, China

Bachelor of Economics and Finance (with Distinction)

Aug. 2021 – June 2025

• **GPA:** 3.93/4.00; **Ranking:** 9/193; **Honors:** Comprehensive Excellence Scholarship (1%), Technological Innovation Scholarship

• **Minor in Statistics:** 4.00/4.00; **CS Coursework:** Deep Learning, Artificial Intelligence, Data Structure, OOP, C++ Programming

University of Pennsylvania

Philadelphia, PA

International Guest Student

Aug. 2023 – Dec. 2023

• **GPA:** 4.00/4.00; **Coursework:** Real Analysis (PhD, A+), Machine Learning (Graduate, A), Analysis of Algorithms, Game Theory

Research Experience

Adaptive Activation-Steered Data Generation for Robust Multi-Turn LLM Guardrails

Pittsburgh, PA

Advisor: Dr. Leman Akoglu, Heinz College, Carnegie Mellon University

July 2026 – Present

- Building an online data-generation loop for LLM safety guardrails, using a bandit controller to select activation-steering configurations and synthesize failure-targeted harmful examples under a fixed budget; manuscript in preparation (**1st Author**)
- Exploring co-training of a multi-turn attack generator with the guardrail to expose evolving blind spots under distributed intent

RelScientist: LLM Agents as Full-Stack Data Scientists for Task-General Relational Learning

Pittsburgh, PA

Advisor: Dr. Leman Akoglu, Heinz College, Carnegie Mellon University

July 2026 – Present

- Developed RelScientist, an LLM agent for relational database learning that engineers SQL features, selects and tunes models, uses cross-run memory, and ensembles predictions; achieved **SOTA** results on RelBench and 4DBInfer against 20 baselines
- Extended RelScientist to **new relational learning tasks** (time-series forecasting, few-label anomaly detection); introduced benchmarks RelBench-TS and RelBench-AD and a task-general agent harness; manuscript in preparation (**1st Author**)

When Less Compute Is More: Adaptive Early-Exit for Tabular Foundation Models

Pittsburgh, PA

Advisor: Dr. Leman Akoglu, Heinz College, Carnegie Mellon University

June 2026 – Sep. 2026

- Found that deeper layers can underperform in anomaly-detection foundation models across 2,200+ datasets and 3 model families, with 4.7–7.3% average AUROC loss; traced the failure to context pollution through mechanistic analysis
- Designed a lightweight dataset-level early-exit router on frozen backbones, **recovering ~45% of the oracle gain with ~1.8× speedup**; achieved **SOTA** results across 3 anomaly detection benchmarks; manuscript submitted to **ICLR 2027 (1st Author)**

Structured Prompt Optimization Meets RL for Global and Local Interpretability over Complex Text

Pittsburgh, PA

Advisor: Dr. Leman Akoglu, Heinz College, Carnegie Mellon University

Sep. 2025 – June 2026

- Designed a 3-stage LLM pipeline (rulebook learning + distillation + RL) producing both global and instance-level explanations for complex text classification across legal, academic and clinical domains; manuscript submitted to **NAACL 2027 (1st Author)**
- Introduced SPO4SOP, a linear-time structured prompt optimizer for interpretable and composable rule sets, and BD-GRPO, a classification-imbalance-robust RL variant; outperformed prompt optimization baselines and end-to-end SFT on both performance and reasoning quality across 3 benchmarks; BD-GRPO reaches peak performance **3-5× faster than vanilla GRPO**

Confidence as Control: A Survey of Confidence Utilization in Large Language Models

Pittsburgh, PA

- Co-authored a survey synthesizing **130+ papers**; proposed a novel taxonomy of LLM confidence scores as an actionable decision signal for training, inference, risk control, model selection, and RAG; manuscript under review at **TMLR (2nd Author)**

Work Experience

Renaissance-Era Investment

Beijing, China

Quantitative Research Intern

June 2023 – Aug. 2023

- Developed alpha strategies for China's A-share market with 10-minute level data; contributed 4 factors for the company
- Trained XGB models and achieved **6%** annual excess returns in out-of-sample, long-only backtests under high transaction costs

Skills

Language: Chinese (Native); English (Fluent, TOEFL: 118/120, GRE: V165+Q170+AW4.0)

Competition: Kaggle Multimodal Single-Cell Integration Competition; Silver Medal as Team Leader (2022)

Technical Skills: Python (PyTorch, Transformers, Scikit-learn, vLLM, VeRL, liteLLM, NumPy, Pandas), C++ (STL), R, LaTeX, Git